National University of Singapore

DSA5205: Data Science for Quantitative Finance

Ridge-Based Monthly Equity Selection

A validation-led comparison and a test of a momentum-screened portfolio

Project 2 Group 45

Member	Full name	Student ID
1	CHEN HONGXING	
2	LIU WEIYE	
3	WANG KE	
4	ZHAO XIAYU	

Monthly decisions Next-open execution 20 bps per traded side

Abstract

We ask whether a momentum–volatility screen followed by Ridge ranking adds investment value after costs. We compare four forecasting models on the same 30 candidates and 15 predictors in 2020–2021. LASSO records the lowest validation MSE and highest Sharpe among the models. Ridge has the highest validation Rank IC, and its L2 penalty combines correlated features. Within Ridge, validation MSE selects $\alpha = 1{,}000$. A supplementary diagnostic finds smaller movements in Ridge's coefficient vector across rolling training windows.

The test covers 48 monthly decisions in 2022–2025. Ridge earns a net CAGR of 43.04% and a Sharpe ratio of 1.272, with $r_f = 0$. Pure momentum within the same screen earns 51.51% and a Sharpe ratio of 1.590. Ridge exceeds the SPY and QQQ buy-and-hold benchmarks in this sample.

Reading the evidence

Forecasts. On the main test, Ridge lowers squared error by 1.87% relative to the pre-test historical mean. The comparison measures predictions for all 30 candidates each month. Section 11 reports the forecast evidence.

Portfolio value. Ridge and B3 trade the same candidate pool at equal weights and face the same cost rule. The difference in their returns measures the effect of changing the ranking signal on that pool. Sections 12 and 15 connect the portfolio results to the research question.

Portfolio extensions. Section 14 examines prediction-based weighting and alternative candidate rules on the same historical dates. The comparisons show the associated changes in return, risk and drawdown.

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1 Research question and market

Can a momentum–volatility screen followed by Ridge ranking add investment value after costs? The primary comparison holds the monthly candidate pool fixed and asks whether Ridge adds value over simple momentum. A broader momentum portfolio and two passive ETFs put the result in context.

Momentum captures the direction and persistence of past returns. Intermediate-horizon continuation motivates a ranking based on 12–1 momentum (Jegadeesh and Titman, 1993). Volatility captures the dispersion of recent returns. Two stocks with similar momentum can reach it through very different price paths. Including several volatility groups lets the model compare stocks with different risk profiles within each momentum group. The screen also limits the chance that the candidate panel represents only one type of price path.

We trade US large-cap equities in US dollars and use the last observed SPY session of each month as the decision date. Figure 1 traces the study from daily data to monthly evaluation. The statistical comparison measures return prediction and ordering across all 30 candidates. The portfolio comparison measures what happens when the ten highest scores are traded.

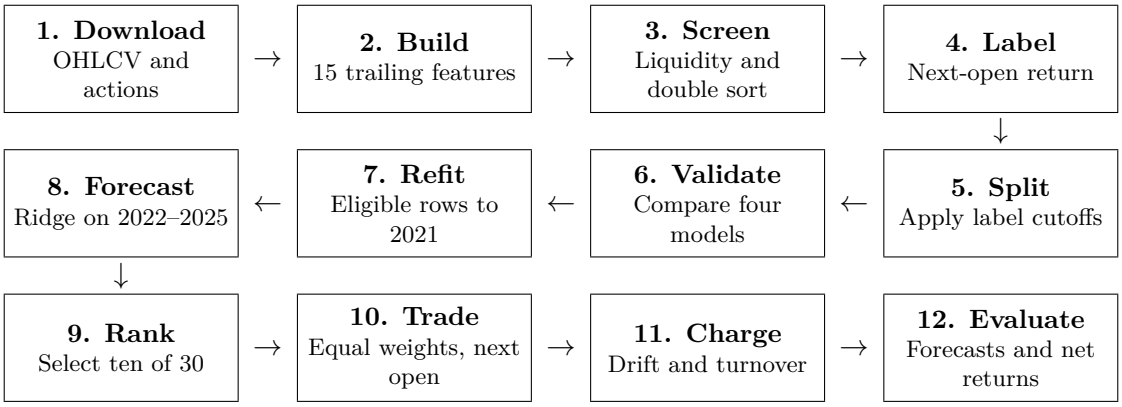


Figure 1: Data-to-decision workflow, adapted from the team's earlier report. Four model families are compared in validation. Ridge forecasts the main test block.

2 Data acquisition and cleaning

We obtain the current S&P 500 ticker list from Wikipedia and download the constituents and the two benchmark ETFs from Yahoo Finance using `yfinance` (Wikipedia, n.d.; Yahoo Finance, n.d.; yfinance, n.d.). The downloader requests daily open, high, low, close and volume, plus the vendor's Adjusted Close, Dividends and Stock Splits fields. It uses `auto_adjust=False` to retain the original OHLC fields and `actions=True` to request action fields. The price history begins in 2009 to supply feature lookbacks for the study's 2010 start.

2.1 Trading calendar, adjustments and raw observations

Dates are normalised to session dates and ticker–date duplicates are reduced to the first record. The pipeline aligns each stock to the observed SPY calendar, keeping unobserved sessions missing. It uses $O_{i,t}^{\text{adj}} = O_{i,t}A_{i,t}/C_{i,t}$, where O is the quoted open, A is Adjusted Close and C is close. The adjustment ratio carries the vendor's dividend and split treatment to the open. The return calculation uses that ratio for stocks and ETFs.

2.2 Invalid and missing values

The adjusted-close series for daily price features treats non-positive values as missing. The 12–1 momentum feature uses observed month-end adjusted closes. Zero denominators in the adjusted-open ratio, beta, volume ratio and intraday range produce missing values. Infinite predictor values become missing before feature filling. The pipeline leaves raw prices, volume and execution prices unfilled. It does not winsorise returns or impose a blanket correction on every raw OHLC field.

Raw coverage is the fraction of sessions with open, close, adjusted close and volume all present. It is measured before feature filling. The screen requires at least 95% coverage over each of 21, 63 and 252 sessions. A missing auxiliary predictor can carry forward from the same stock for up to two sessions. The three screening predictors `mom_2_12`, `vol_63` and `ln_adv_20` are never filled. This keeps liquidity eligibility and the sorting variables tied to their contemporaneous trailing estimates. Candidate rows require all predictors to be finite. The final modelling panel also requires finite stock and ETF holding-period returns.

The fill allowance applies to the 21-, 63- and 126-session momentum measures, relative momentum, 21-session volatility, beta, drawdown, both moving-average ratios, RSI, the volume ratio and the intraday range. Table 3 gives their field names.

2.3 Historical universe

Backfilling today's constituent list can omit stocks that left the index and include later entrants before they joined. This creates survivorship and membership look-ahead concerns for the active stock portfolios. SPY and QQQ use historical ETF prices and returns. Their historical holdings are not reconstructed from the current list. The submitted derived panel fixes the stock snapshot for reproducibility, while a fresh download can change that snapshot.

3 Candidate screen and fixed strategy design

The screen starts with stocks that close at or above \$5, have at least \$10 million of average daily dollar volume over 20 sessions, meet the raw coverage requirements and have finite values for all predictors. ADV20 averages close times volume over 20 sessions and requires 19 observations. The \$10 million floor excludes thinly traded names. Keeping up to 300 eligible stocks by ADV20 gives the monthly portfolio a broad pool while focusing on names with higher observed trading activity. Liquidity is a screening proxy and the backtest applies a separate cost assumption.

The 3×3 sort is dependent. We first split the liquid universe into high, middle and low momentum groups using 12–1 momentum. We then rank 63-session realised volatility independently within each momentum group and split it into three approximately equal cells. The high-momentum group takes the strongest names within each cell, the low group the weakest and the middle group the names closest to the cell's median momentum. Ticker order breaks ties. Table 1 gives the quotas, which produce ten names per momentum group and 30 in total.

Momentum group	Low volatility	Middle volatility	High volatility	Total
High	3	4	3	10
Middle	3	4	3	10
Low	3	4	3	10
Total	9	12	9	30

Table 1: Fixed cell quotas for the monthly candidate panel.

The model ranks all 30 candidates and holds the ten highest scores. Ten names give the ranking room to be selective while spreading the portfolio across several stocks. Equal 10% target weights isolate the value of selecting the highest scores from the separate question of sizing positions with forecasts. The Top 300 limit, 3×3 sort, 3–4–3 quotas and Top 10 equal-weight rule are fixed design choices in the primary pipeline. The validation search varies model settings. The main test applies the fixed portfolio rule.

4 Benchmark design

Table 2 defines the four validation portfolios and four fixed benchmarks. All model portfolios apply the same trade rule so their validation comparison reflects differences in scores.

Portfolio	Role	Selection rule	Weighting
Ridge	Candidate model	Top 10 Ridge scores in the screened pool	Equal 10%
LASSO	Candidate model	Top 10 LASSO scores in the screened pool	Equal 10%
Random Forest	Candidate model	Top 10 Random Forest scores in the screened pool	Equal 10%
XGBoost	Candidate model	Top 10 XGBoost scores in the screened pool	Equal 10%
B1	Momentum benchmark	Top 10 by 12–1 momentum in the full liquid pool	Equal 10%
B3	Momentum benchmark	Top 10 by 12–1 momentum in the screened pool	Equal 10%
SPY	ETF benchmark	Buy and hold the S&P 500 ETF	Fully invested
QQQ	ETF benchmark	Buy and hold the Nasdaq-100 ETF	Fully invested

Table 2: Portfolio rules for the four validation models and four benchmarks.

B3, ranking value. B3 ranks the same 30 names by 12–1 momentum and uses the same weights, dates and fees. Its comparison with Ridge isolates the choice of ranking signal in the screened pool.

B1, the opportunity set. B1 selects by the same momentum score from up to 300 liquid names. Its comparison with B3 shows how the double sort changes the momentum portfolio. Its comparison with Ridge spans both the candidate set and the ranking signal.

SPY and QQQ, passive alternatives. SPY tracks the S&P 500 and represents broad US large-cap exposure. QQQ tracks the Nasdaq-100 and gives a second reference with a different sector and growth exposure. Both use the same holding interval as the strategy and incur the initial entry cost. We use a pre-test historical mean as the separate baseline for forecast R^2.

5 Target, features and preprocessing

For a stock selected at decision date t, the label is the adjusted-open simple return

$$y_{i,t} = \frac{P^{\text{adj,open}}_{i,\text{next rebalance}}}{P^{\text{adj,open}}_{i,\text{entry}}} - 1.$$

Figure 2 shows how features, execution and return realisation align. Each row records the decision, execution and label-end dates. The last date governs fitting eligibility in Section 8.

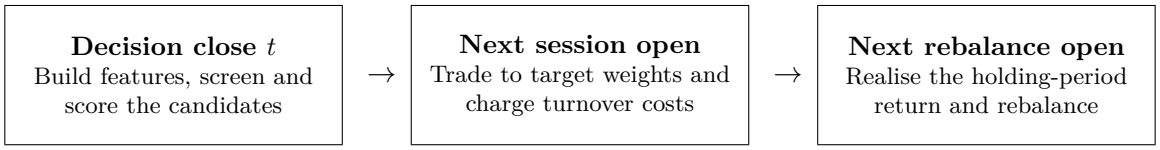


Figure 2: Decision, execution and label alignment for one monthly holding period.

Table 3 lists the 15 trailing features and their economic interpretation. Rolling volatility uses at least 20 of 21 or 60 of 63 daily returns, and beta uses at least 240 paired returns in 252 sessions. The two moving averages require 95% of their lookback observations.

Group	Feature	Definition and lookback	Interpretation
Momentum	mom_21	Adjusted close / close 21 sessions ago, less one	Near-term return signal
Momentum	mom_63	Adjusted close / close 63 sessions ago, less one	Quarterly momentum
Momentum	mom_126	Adjusted close / close 126 sessions ago, less one	Semi-annual momentum
Momentum	mom_2_12	Adjusted month-end close at $m-1$ / at $m-12$, less one	First sorting signal
Momentum	relative_mom_63	63-session return minus SPY return	Relative strength
Risk	vol_21	21-session daily return standard deviation $\times\sqrt{252}$	Short-term risk
Risk	vol_63	Same volatility measure over 63 sessions	Risk and second sort
Risk	beta_252	252-session paired return covariance with SPY / SPY variance	Systematic exposure
Risk	drawdown_63	Adjusted close / 63-session maximum adjusted close, less one	Recent loss state
Trend	ma_ratio_50	Adjusted close / 50-session mean, less one	Short-term trend
Trend	ma_ratio_200	Adjusted close / 200-session mean, less one	Long-term trend
Trend	rsi_14	$100-100/(1+G/L)$, with 14-session average price gains G and losses L	Recent directional pressure
Liquidity	ln_adv_20	$\log(1+\text{ADV20})$ for eligible stocks	Size and liquidity
Liquidity	volume_ratio_21	Volume relative to 21-session average	Abnormal activity
Range	range_hl	Current session's (high minus low) / close	Intraday price pressure

Table 3: The fifteen predictors. Price-return numerators and denominators use adjusted closes.

The 12–1 signal separates medium-horizon continuation from the most recent month. Risk features describe dispersion, market exposure and recent losses. Trend, liquidity and intraday range describe the path and trading conditions behind observed returns.

6 Models and control of overfitting

The 15 features fall into five economic groups. Several measure related effects at different horizons. For example, 63-session momentum and its market-relative version have a training correlation of 0.939. The correlation between 126-session momentum and the 200-session moving-average ratio is 0.936. Appendix A reports the training-period correlations.

Ridge gives the related predictors a joint linear score. It fits an intercept b and penalises the squared coefficient norm (Hoerl and Kennard, 1970)

$$(\hat{b}, \hat{\beta}) = \arg\min_{b,\beta} \left\{ \sum_i (y_i - b - x_i^\top \beta)^2 + \alpha \sum_j \beta_j^2 \right\}.$$

We use an additive score with the same fitted weights across stocks and months. With correlated predictors, ordinary least squares can assign very different weights to two similar signals with little change in training error. L2 shrinkage trades some bias for lower estimation variance and lets related predictors contribute together. This is a reasonable starting point when many signals may carry small amounts of information.

LASSO applies an L1 penalty, which can set coefficients to zero and tests whether a smaller set of signals is enough (Tibshirani, 1996). Random Forest averages regression trees fitted to bootstrap samples and considers a random subset of predictors at each split (Breiman, 2001). XGBoost adds trees sequentially to reduce a regularised squared-error objective (Chen and Guestrin, 2016). The trees can capture interactions, such as a different relationship between momentum and return at high versus low volatility.

The regularised linear models give a direct comparison of sparse and dense signals. The tree models serve as checks for nonlinear gains on the same inputs. We limit flexibility with regularisation, tree depth and leaf-size controls, then compare a small grid on later dates. This combination targets both estimation noise in the models and chance improvements from searching many settings.

7 Training methods and model settings

We fit all 23 configurations in Table 4 on the same training observations. The reference run uses Python 3.12, scikit-learn 1.9.1 and XGBoost 3.2.0. The runnable environment and remaining library versions are documented in the README.

Model	Grid	Count
Ridge	$\alpha \in \{0.1, 1, 10, 100, 1000, 10000\}$	6
LASSO	$\alpha \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$	5
Random Forest	Depth 3 or 6, minimum leaf size 10 or 25	4
XGBoost	Depth 2 or 3, learning rate 0.03 or 0.10, 100 or 200 trees	8

Table 4: The 23 configurations evaluated on validation.

Ridge and LASSO. Both fit an intercept and standardise each predictor using means and scales estimated inside the training pipeline. Ridge uses scikit-learn's automatic solver selection, which resolves to Cholesky on our dense training matrix. LASSO uses cyclic coordinate descent with a 20,000-iteration cap and a tolerance of 10^{-4}.

Random Forest and XGBoost. Both use unscaled inputs. Random Forest fits 250 bootstrap trees with squared-error splits and a feature fraction of 0.7. XGBoost uses the histogram tree method, squared-error loss, row and column fractions of 0.8, and an L2 tree penalty of 10. The XGBoost tree count is chosen from the grid. The estimator setup passes seed 42 to LASSO and the two tree methods. Ridge's dense solver is deterministic.

8 Chronological split and fitting

The raw price history starts in 2009, which supplies lookback observations for 2010 decisions. We fixed the training, validation and test date blocks in Table 5 before fitting the models. A row enters a fitting or validation block when its holding-period return has settled by the corresponding cutoff. For example, the 30 November 2021 signal trades at the December open and exits on 3 January 2022. Its return is unavailable at the 1 January cutoff, so this decision is excluded from validation and the final refit. The same cutoff logic removes the last two decisions of 2019 from initial training and the last two decisions of 2021 from validation and refitting.

Use	Decision years	Months	Rows	Last label end
Train each configuration	2010–2019	118	3,540	2 Dec 2019
Compare configurations	2020–2021	22	660	1 Dec 2021
Refit selected model	2010–2021	142	4,260	1 Dec 2021
Test selected model	2022–2025	48	1,440	2 Feb 2026

Table 5: Eligible observations after applying the realised-label cutoff. Each decision month contains 30 candidates.

An eligible label ends on or before its block cutoff. The chronological split preserves the forecast order and the label embargo keeps future exits out of each fit. The scaler for each linear model is fitted on the eligible training rows and applied to the later validation observations.

For the main test, we refit Ridge and its scaler on the 4,260 eligible pre-test rows, including 2019 decisions whose labels had settled by the later refit cutoff. The refit uses the selected α and stays fixed across the 48 test decisions.

9 Validation and choice of Ridge

We use validation MSE to choose a setting within each model family. Table 6 compares the selected configurations and benchmarks on the same 22 months. All four models use the same 30 candidates and equal-weight Top-10 rule. MSE measures error in return levels. Rank IC averages the monthly Spearman correlation between predicted and realised ranks across all 30 candidates. Validation Sharpe uses net monthly returns and $r_f = 0$.

Strategy	MSE	Rank IC	Net CAGR	Sharpe
Ridge	0.022422	**0.0438**	45.69%	0.918
LASSO	**0.022281**	0.0250	46.14%	0.926
Random Forest	0.022412	0.0261	40.20%	0.866
XGBoost	0.022642	-0.0098	26.59%	0.693
B1	—	—	64.13%	1.341
B3	—	0.0332	38.41%	1.074
SPY	—	—	23.17%	1.048
QQQ	—	—	38.72%	1.566

Table 6: Post-cost validation results over the same 22 monthly decisions. Sharpe uses a zero risk-free rate.

LASSO has the lowest MSE (0.022281) and highest Sharpe (0.926) among the four models. Its selected fit sets six of 15 coefficients to zero. Ridge has the highest Rank IC (0.0438 versus 0.0250 for LASSO), with MSE 0.63% higher and a slightly lower Sharpe of 0.918. Random Forest and XGBoost record Rank ICs of 0.0261 and -0.0098, and their Sharpe ratios trail both linear models. We use Ridge as the primary specification because the strategy trades the ordering and L2 shrinkage retains related predictors. Rank IC covers all candidates, so the Top-10 portfolio still requires a separate economic test.

Ridge α	0.1	1	10	100	1,000	10,000
Validation MSE	0.022584	0.022582	0.022567	0.022481	**0.022422**	0.022784
Validation Rank IC	0.0361	0.0369	0.0357	0.0421	0.0438	0.0560

Table 7: Ridge validation grid. Bold indicates the minimum MSE used to select the penalty.

Table 7 shows that stronger shrinkage helps validation MSE up to $\alpha = 1{,}000$. The largest penalty improves the ordering but raises squared error. The within-family MSE rule therefore selects $\alpha = 1{,}000$ for the final refit and test.

The primary specification and its penalty use validation evidence and the model rationale. Test-period metrics do not enter those choices. The project had already reviewed some out-of-sample results during development, and the stability diagnostic was added later using the training block.

9.1 Coefficient stability across training windows

The supplementary diagnostic fits Ridge and LASSO on 59 rolling windows of 60 months within the training block, advancing one month at a time. Each fit standardises its own training features. We compare adjacent coefficient vectors using

$$\frac{\|\hat{\beta}_w - \hat{\beta}_{w-1}\|_2}{\|\hat{\beta}_{w-1}\|_2},$$

the size of the change relative to the previous coefficient vector. This measures how much the estimated weights move together as the training window shifts.

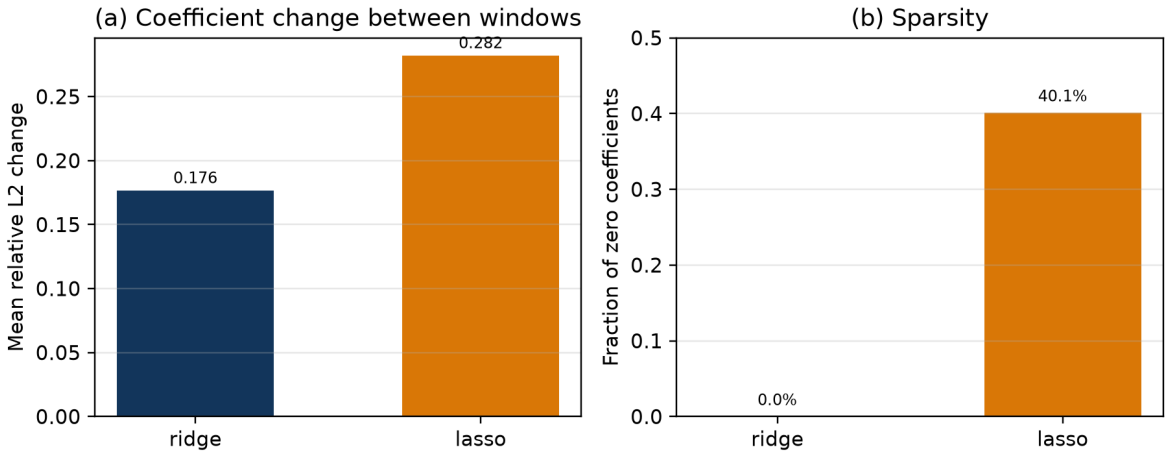


Figure 3: Rolling training fits. The left panel averages relative coefficient changes across adjacent windows. The right panel shows the share of fitted coefficients equal to zero.

Figure 3 summarises movement and sparsity. Table 8 shows that Ridge's mean relative change is 37.4% lower. It records more direct sign changes, which count a positive-to-negative or negative-to-positive switch regardless of size. The smaller movement in Ridge's full vector supports dense shrinkage for these inputs. Portfolio turnover depends on how scores and holdings change each month.

Metric	Ridge	LASSO
Mean relative coefficient change	0.176	0.282
Zero coefficients across all windows	0.0%	40.1%
Direct sign changes across adjacent windows	50	1

Table 8: Coefficient stability over the same 59 rolling training windows.

10 From forecasts to trades

Figure 4 shows the path from candidates to trades. We rank the 30 stocks by predicted return at the decision close, break score ties by ticker and set the remaining targets to zero. The portfolio is fully invested, long-only and unlevered. In Figure 5, TGT ranks tenth with a 1.97% score and receives 10%. CARR ranks eleventh at 1.86% and receives zero. Orders execute on 1 July 2025.

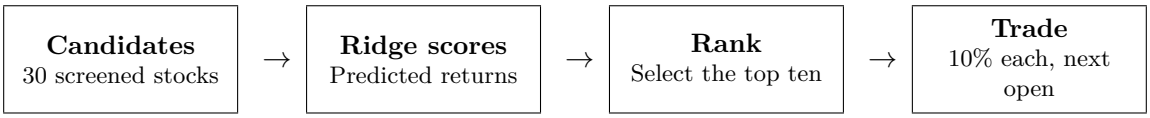


Figure 4: Forecast-to-trade mapping for Main Ridge.

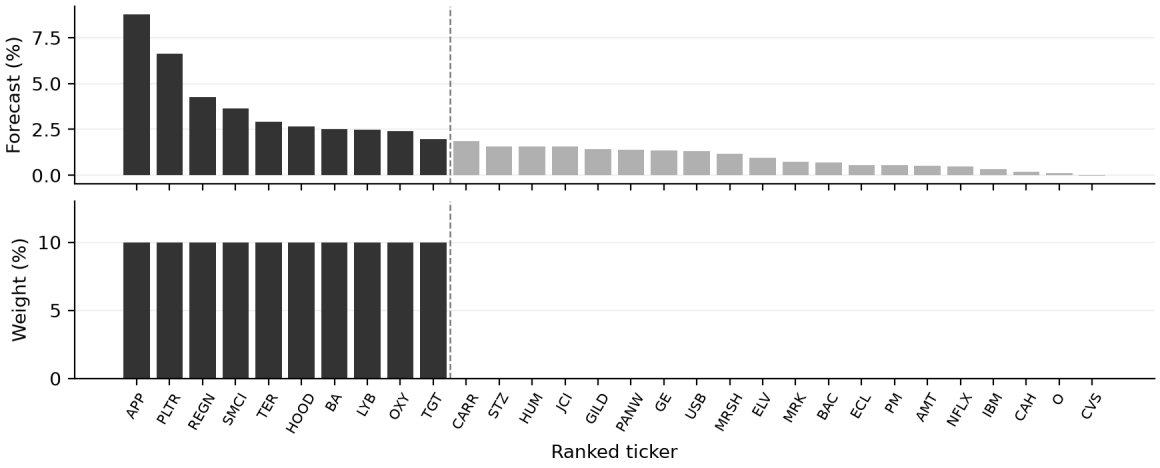


Figure 5: Ridge scores and equal weights for the 30 June 2025 decision.

Weights drift with stock returns before the next rebalance. If $w_{i,t}^{\text{target}}$ and $w_{i,t}^{\text{pre}}$ are the new and drifted weights, the fee is

$$c_t = 0.002 \sum_i |w_{i,t}^{\text{target}} - w_{i,t}^{\text{pre}}|.$$

We subtract it from the gross holding-period return. Buys and sells each cost 20 bps per dollar traded. Replacing one 10% holding with another at its target weight costs 4 bps of portfolio value if the sold position is also worth 10% at the trade. B1 and B3 use the same cost rule. SPY and QQQ each incur a single entry fee.

11 Ridge forecasts on the test period

The test spans 48 monthly decisions from 31 January 2022 to 31 December 2025. Actual holding starts on 1 February 2022 and ends on 2 February 2026. Consecutive holding periods meet at the next rebalance open.

Forecast metric	Ridge
Observations	1,440
MSE	0.015057
MAE	0.082802
Mean monthly Rank IC	0.0619
Months with positive Rank IC	56.25%
Out-of-sample R^2	0.0187

Table 9: Forecast performance on the 30-stock test panel.

Table 9 reports the prediction metrics. For the benchmark-relative R^2, we compare Ridge with the constant mean return from the eligible pre-test refit sample (Campbell and Thompson, 2008). The score is $1 - \mathrm{SSE}_{\mathrm{Ridge}}/\mathrm{SSE}_{\mathrm{mean}}$. Figure 6 shows a 1.87% reduction in squared error relative to that forecast. The mean Rank IC of 0.0619 indicates a modest positive association between predicted and realised ranks. Both measures use all 30 candidates, so improvements in the middle of the ranking also contribute. The portfolio result depends on which names enter the top ten and how much it costs to trade them.

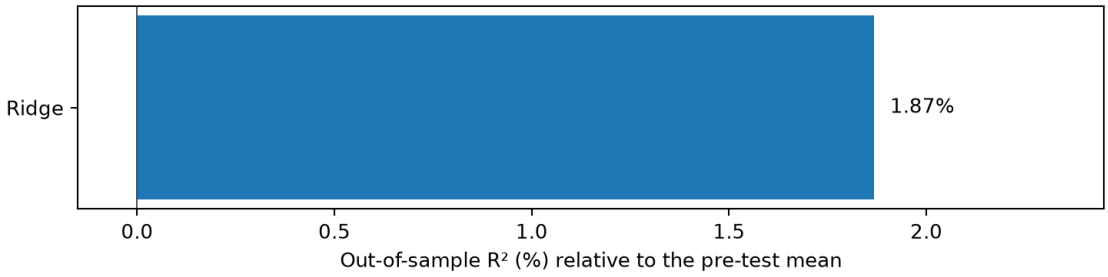


Figure 6: Test-period forecast improvement against the historical-mean predictor.

12 Ridge portfolio and benchmark results

Table 10 reports net CAGR, annualised monthly volatility and Sharpe, drawdown, and two-way turnover. We compound monthly net returns to form wealth. CAGR annualises the final wealth over 48 months. Volatility and the mean-to-standard-deviation Sharpe use a $\sqrt{12}$ multiplier, with $r_f = 0$ for Sharpe (Sharpe, 1966). Drawdown measures the largest decline from the running peak of rebalance-period wealth.

Strategy	Net CAGR	Volatility	Sharpe	Max drawdown	Turnover
Ridge	43.04%	32.56%	1.272	-28.54%	1.26
B1	73.13%	37.72%	1.664	-19.19%	0.65
B3	51.51%	29.04%	1.590	-14.99%	0.94
SPY	12.72%	15.27%	0.863	-19.67%	0.02
QQQ	14.81%	19.93%	0.796	-25.56%	0.02

Table 10: Post-cost test results for Ridge and the four fixed benchmarks. CAGR, volatility and Sharpe are annualised. Turnover is the average monthly sum of absolute weight changes.

Figure 7 shows that one dollar grows to \$4.19 with Ridge, \$1.61 with SPY and \$1.74 with QQQ. Ridge earns 43.04% net CAGR and 1.272 Sharpe, ahead of both ETFs. It also has higher volatility (32.56% versus 15.27% and 19.93%) and a deeper drawdown (28.54% versus 19.67% and 25.56%).

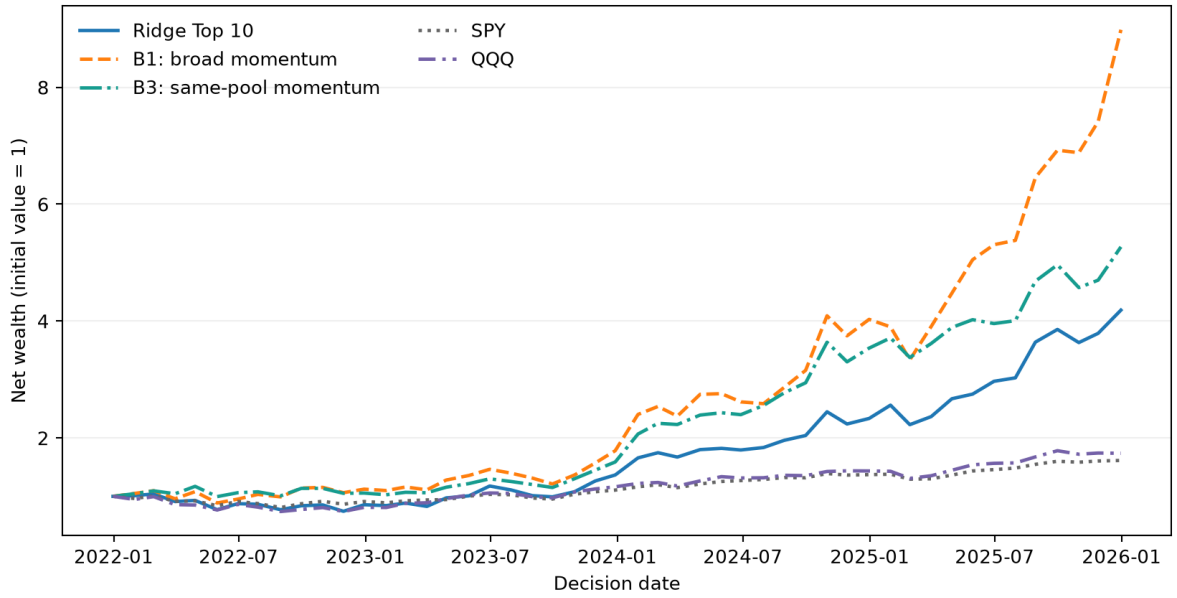


Figure 7: Cumulative net wealth, indexed to the decisions that initiated each holding period.

B3 leads Ridge by 7.53 percentage points of gross CAGR and by 8.47 after costs. Ridge's higher turnover widens that gap by 0.94 points. B1 has the highest net CAGR (73.13%) and the lowest turnover among the active portfolios (0.65). It also has higher volatility than Ridge. Section 15 discusses what these comparisons reveal about the screen and ranking.

13 Performance by decision year

Table 11 groups net returns by decision year, with 12 decisions per cohort. A December decision trades in January of the following year. Figure 8 plots those returns alongside Sharpe estimates from each cohort's monthly returns.

Decision year	Ridge	B1	B3	SPY	QQQ
2022	-14.36%	12.33%	5.63%	-8.82%	-18.79%
2023	59.52%	58.61%	50.35%	21.42%	43.23%
2024	70.77%	126.14%	122.68%	23.86%	23.34%
2025	79.47%	122.99%	49.00%	17.72%	21.10%

Table 11: Net compound returns by decision year.

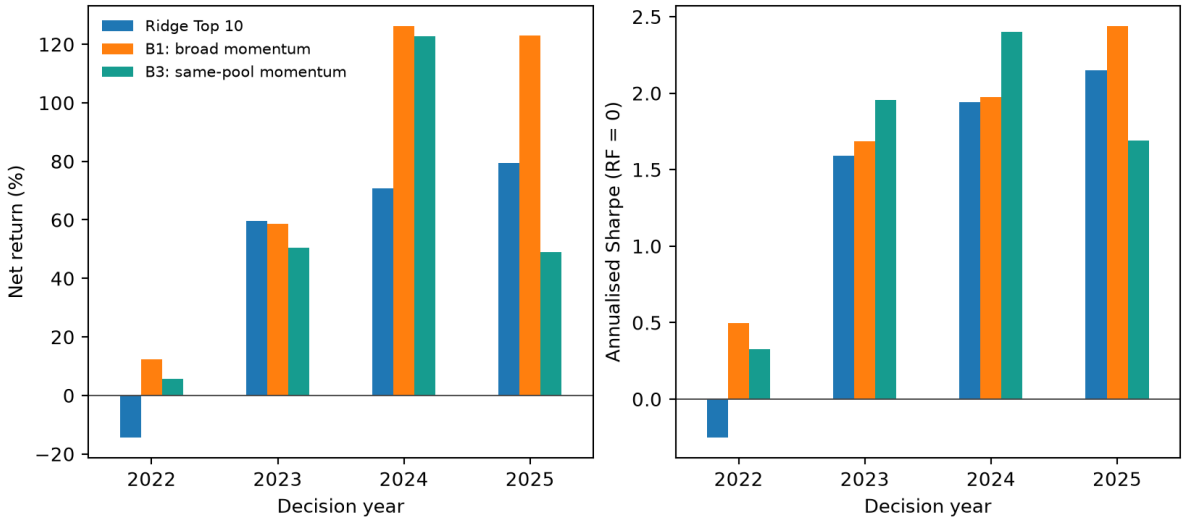


Figure 8: Net returns and Sharpe ratios by decision-year cohort.

Ridge earns a higher net return than B3 in 2023 and 2025. Its higher return in 2023 comes with a lower Sharpe. In 2025 it leads B3 on both measures. B3 has the higher Sharpe in three of the four years. Its largest return advantage is in 2024, when B3 earns 122.68% and Ridge earns 70.77%. In 2022, Ridge loses 14.36% while B3 gains 5.63%. B1 earns 126.14% in 2024 and 122.99% in 2025, when the broad momentum portfolio builds much of its lead.

14 Exploratory Ridge portfolio extensions

After reviewing earlier test results, we examined prediction-based weighting, a broader candidate set, and momentum-group quotas. These exploratory variants reuse the 48 test decisions and 20 bps per traded side. Their Ridge fit uses 60 candidates per month, with 20 drawn from each momentum group. Its own validation MSE selects $\alpha = 1{,}000$. The 30-candidate, equal-weight rule is represented by Main Ridge in Table 10.

Rule	Portfolio construction	Net CAGR	Vol.	Sharpe	Max DD
B1	Broad momentum Top 10, equal weight	73.13%	37.72%	1.664	−19.19%
B2	Broad momentum Top 30, equal weight	31.40%	25.85%	1.193	−17.33%
B3	Screened momentum Top 10, equal weight	51.51%	29.04%	1.590	−14.99%
R1	B1 stock list, Ridge score weights capped at 20%	87.89%	44.93%	1.642	−21.47%
R2	Ridge Top 10 from 60 candidates, equal weight	37.67%	39.86%	1.005	−46.63%
R3	Ridge Top 3/4/3 by momentum group, equal weight	22.53%	33.26%	0.778	−38.99%

Table 12: Exploratory portfolios and momentum comparators, net of costs. Sharpe uses $r_f = 0$. Maximum drawdown uses rebalance-period wealth.

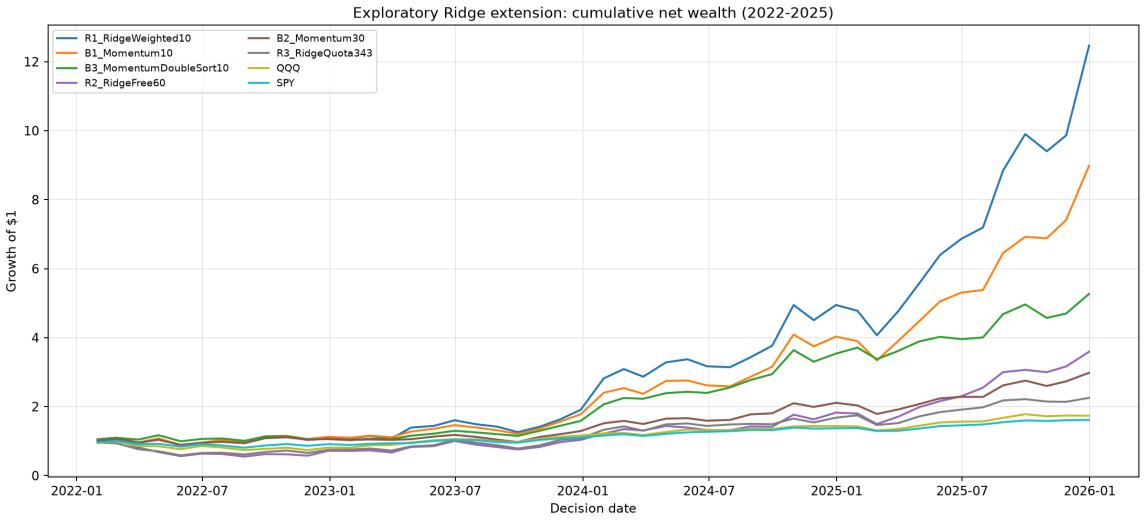


Figure 9: Cumulative net wealth for the exploratory variants, momentum comparators and ETFs.

Table 12 and Figure 9 show the trade-offs. R1 weights B1's ten names using positive Ridge scores. Excess weight above 20% is redistributed, and an all-zero allocation falls back to equal weights. Net CAGR rises from 73.13% to 87.89%, volatility rises from 37.72% to 44.93%, and Sharpe falls from 1.664 to 1.642. R2 ranks the 60-name pool globally. R3 takes three, four and three names from the high, middle and low momentum groups. This quota reduces drawdown from 46.63% to 38.99%, with lower return and Sharpe. The results are exploratory because the sample had already been examined.

15 Interpretation and limitations

Answer to the research question. The screened Ridge portfolio beats both passive ETFs on net CAGR and Sharpe in this sample, with higher risk. The matched B3 portfolio beats Ridge on both return and risk-adjusted performance. The evidence supports useful return predictions from Ridge and favours simple momentum for ranking within this screen.

Why momentum led. The observed B3 advantage predates costs. Ridge's higher turnover then makes the net gap larger. Ridge fits return levels and its Rank IC covers the entire candidate set. Small score changes around rank ten can alter the holdings, and the fitted weights may pick up common market variation or noise that adds little at that cutoff. The data show the performance and cost differences. These mechanisms are possible explanations.

B1 and B3 use the same momentum signal. B1's lead is therefore associated with applying momentum to a broader pool. One possible explanation is that the cell quotas removed some of the strongest names available to B1. This comparison measures the effect of the full screen on this period, including the volatility sort and fixed quotas.

Market exposure. A regression on SPY using 48 monthly observations gives Ridge a beta of 1.64 and an annualised single-factor intercept of 21.73%, with a zero risk-free rate. The beta indicates greater sensitivity to market moves than holding SPY directly. The intercept is the return component left after adjusting for that one exposure and can include other effects, such as momentum. The same-pool B3 comparison is the more direct measure of what Ridge added to the ranking rule.

Stock universe. The current-constituent sample can favour stocks that survived or grew enough to enter the index. Ridge and B3 share that pool, so the comparison controls for the candidate list in each month. Both active portfolios remain conditional on that list. The historical ETF series provide passive market context and are unaffected by our stock membership screen.

What we learned. Forecast MSE, ranking and net portfolio performance answer different questions. The four-year result supports keeping the matched momentum rule as the test of incremental value. Relative performance varies across years, and the fixed cost assumption leaves out stock-specific spreads and market impact.

16 Reproducibility and contributions

The monthly study uses the packaged derived panels and the following commands from the project root.

```
python -m pip install -r requirements.txt
python run_primary_study.py
```

The notebook runs the same workflow. The `--rebuild-data` flag regenerates the monthly panels from Yahoo Finance and the current constituent list. To regenerate the exploratory Ridge results from the packaged candidate panel, run `python -m src.ridge_extension`. Table 13 maps each output to the evidence in this report.

Location	Contents
`config/experiment_config.yaml`	Dates, seed, model grids and portfolio settings.
`results/validation/`	Validation model and benchmark results, tuning grid and Ridge decision record.
`results/oos/`	Ridge forecasts, benchmark returns, annual results and performance table.
`results/diagnostics/`	Rolling training coefficients and stability summary.
`results/extensions/`	Monthly returns and summaries for the exploratory Ridge variants.
`figures/`	Forecast, portfolio and diagnostic figures.

Table 13: Reproduction map.

Member contributions

Table 14 records the work by team member. The title page lists the Group 45 roster.

Member	Contribution
CHEN HONGXING	Proposed the trading strategy, led the candidate screening and stock selection design, and synthesised the benchmark findings.
LIU WEIYE	Prepared the market data, constructed features and forward-return labels, and checked the alignment of decision, execution and label dates.
WANG KE	Trained and validated the four models, compared hyperparameter settings, and documented the choice of Ridge.
ZHAO XIAYU	Ran the backtests with transaction costs, compared performance against the benchmarks, produced the figures and organised the submission files.

Table 14: Team contributions.

AI use

The team used generative AI for code organisation, diagnostic and consistency checks, drafting, and report editing. The reported results come from the project code and saved data and output files. The team is responsible for the research design and interpretation.

A Training-period diagnostics

Twelve of 105 feature pairs have a training-sample absolute correlation of at least 0.70. Figure 10 illustrates how momentum and trend overlap, and how volatility measures move together.

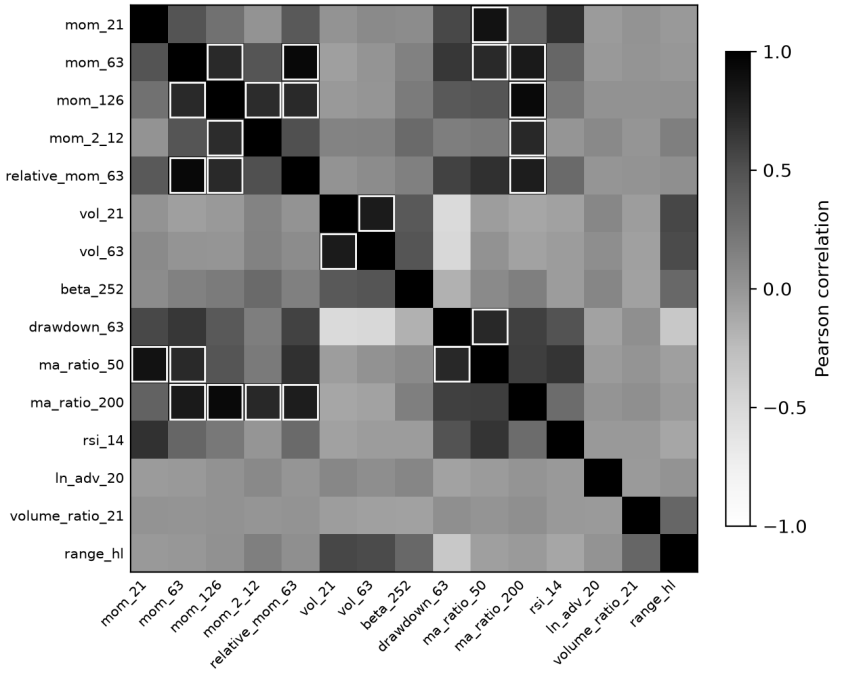


Figure 10: Pearson correlations of the 15 features on the training rows.

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